NIKSHITHA VADTHYAVATH

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EDUCATION

Bachelor of Technology in Information Technology **2022–2026**
Indian Institute of Information Technology Sonepat, Haryana **CGPA: 8.0/10**

EXPERIENCE

- **Founder & CEO** *Jan 2026 - Present*
  Gozy Private Limited
  – Built responsive web application using React frontend with Material-UI components; engineered interactive dashboard for monitoring content pipeline metrics and user engagement.
  – Developed careers page with dynamic job listings; integrated backend APIs for job posting, candidate filtering, and application management with JWT authentication.
  – Developed mobile app using React Native enabling cross-platform accessibility; coordinated API integration with backend services for seamless frontend-backend data flow.
  – Managed startup operations including product strategy, vendor coordination, and client relationship management; led go-to-market positioning and enterprise client acquisition.

- **Research Intern** *Certificate*
  IIT Hyderabad — Federated Learning for Emotion Recognition Jun 2025 – Aug 2025
  – Implemented FedProto federated learning algorithm on FERPlus emotion recognition dataset with 3 distributed clients: Client 1 (happiness, sadness, neutral), Client 2 (disgust, anger, contempt), Client 3 (surprise, fear).
  – Curated and annotated 5,000+ facial emotion samples from DeepFace Emotions dataset; implemented stratified sampling and PyTorch data loaders for balanced non-IID distribution across clients.
  – Trained CNN model over 225 communication rounds using PyTorch with differential privacy constraints
  – Evaluated model performance using Scikit-Learn metrics; visualized convergence curves and per-class performance.

- **LLM Post-Training Intern** *Certificate*
  Ethara.ai — Multimodal AI Alignment Feb 2026 – Apr 2026
  – Annotated and evaluated LLM responses across text, image, audio, and video modalities; designed evaluation rubrics ensuring consistency across 1,000+ samples.
  – Conducted prompt engineering experiments testing 50+ variations (zero-shot, few-shot, chain-of-thought); analyzed LLM output patterns using structured evaluation frameworks.
  – Performed ELO-based comparative benchmarking to evaluate model outputs; documented performance deltas and identified systematic failure modes.
  – Contributed to multimodal alignment dataset combining text and visual reasoning tasks; maintained data validation pipeline for quality assurance.

PROJECTS

- **Endangered Species Detection & Prevention** *Nov 2024 – Apr 2025*
  React.js, Flask, PyTorch, YOLO, OpenCV | GitHub | Live Demo
  – Developed a full-stack wildlife conservation platform using React.js and Flask for endangered species identification.
  – Integrated YOLO-based object detection to identify species from user-uploaded images and provide conservation insights.
  – Built a knowledge retrieval module displaying IUCN Red List information, habitat details, and conservation status.
  – Deployed the application on Netlify with a responsive user interface and real-time prediction workflow.

- **Grammar Scoring Engine** *Apr 2025 – May 2025*
  Streamlit, Python, SpeechRecognition, TextBlob, LanguageTool, NLP | GitHub
  – Developed an AI-powered grammar evaluation application using Streamlit, SpeechRecognition, and Natural Language Processing.
  – Implemented speech-to-text transcription from recorded audio and analyzed grammatical correctness of generated transcripts.
  – Integrated TextBlob and LanguageTool to detect grammatical errors and generate automated feedback with grammar scores.
  – Designed an interactive web interface supporting both audio recording and file uploads for spoken English assessment.

PUBLICATIONS

- Nikshitha, V., et al. "AI-Driven Endangered Species Detection and Conservation Using Deep Learning." *2026 IEEE CIPHER*, NIT Jalandhar, Feb 2026. DOI: 10.1109/CIPHER70417.2026.11523743. [IEEE Xplore]
- Nikshitha, V., et al. "Dual-Stage Deep Learning Model for Glaucoma Detection through Optic Disc–Cup Segmentation and CDR Analysis." *2026 IEEE CIPHER*, NIT Jalandhar, Feb 2026. DOI: 10.1109/CIPHER70417.2026.11523802. [IEEE Xplore]

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, HTML, CSS, SQL, Java
- **ML/AI & Deep Learning:** Machine Learning, PyTorch, TensorFlow, Scikit-Learn, CNN, Transfer Learning, Federated Learning, RAG, LangGraph, Transformers, Computer Vision, NLP, Speech Recognition
- **Full Stack Development:** MERN Stack, React, React Native, Node.js, Express.js, Flask, Streamlit, REST APIs, PostgreSQL, JWT Authentication, Docker
- **Tools & Frameworks:** Git, Jupyter, OpenCV, Plotly, Matplotlib, Seaborn, AWS (EC2, S3), Netlify, Vercel